

Research Paper 02 Summary

The Collection 6 MODIS Active Fire Detection Algorithm

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Giglio et al. (2016), *Remote Sensing of Environment*

1 Basic Information

- **Paper Title:** The Collection 6 MODIS Active Fire Detection Algorithm and Fire Products
- **Authors:** Louis Giglio, Wilfrid Schroeder, and Christopher O. Justice (2016)
- **Sensor Covered:** MODIS (Moderate Resolution Imaging Spectroradiometer) onboard NASA's Terra and Aqua satellites (1 km resolution).

2 What is this Paper About? (In Simple Words)

MODIS is NASA's longest-running global fire monitoring instrument, orbiting Earth since the year 2000. This paper explains the updated "Collection 6" computer algorithm that NASA uses to find fires in MODIS satellite images.

- **How it finds fires:** The algorithm scans satellite pictures in square grids ($1\text{ km} \times 1\text{ km}$). It uses dynamic "windows" (expanding from 3×3 up to 21×21 neighboring pixels) to compare a candidate hot pixel with its surroundings.
- **Filtering false alerts:** It includes built-in filters to reject things that look hot but are not fires, such as sunlight reflecting off water (sun glint), desert sand heated by the sun, and bright rooftops.
- **Fire Radiative Power (FRP):** It explains how the satellite measures the strength and energy output of the fire in megawatts (MW).

3 The Core Limitation: Rigid Rules vs. Complex Reality

The MODIS algorithm relies on a complex tree of fixed, hand-crafted mathematical rules (if-else threshold tests):

- While these static rules are great at filtering out sunlight reflections, they cannot understand land use or human activity.
- As noted in the paper's section on false alarms, stationary human heat sources (such as steel mills, refineries, and factory furnaces) regularly cross the threshold and get permanently tagged as wildland fires.

4 How This Helps Our Project (PS26162)

4.1 1. Providing 20+ Years of Historical Training Data

While newer VIIRS satellites give sharper images (375 meters), MODIS provides continuous data from 2000 to the present (over two decades).

- **Historical Baselines for Factories:** We can look back 10 to 20 years at major Indian industrial centers (like Jamnagar refineries or Tata Steel in Jamshedpur) to see their normal, long-term heat patterns.
- **Training AI Models:** This massive archive allows us to train our AI model on thousands of past seasonal cycles across India.

4.2 2. Justifying Why an AI Model is Better Than Fixed Rules

This paper demonstrates the limits of traditional rule-based algorithms. Hand-crafted ‘if-else’ formulas cannot account for the variety of industrial operations, seasonal crop burning, and forest fires across India. Our AI model (using machine learning) directly improves upon this by learning complex patterns automatically.

4.3 3. Ingesting Multi-Satellite Feeds in Our GIS Backend

NASA FIRMS delivers fire alerts from both MODIS and VIIRS satellites.

- Terra passes over in the morning (~10:30 AM/PM).
- Aqua passes over in the afternoon (~1:30 PM/AM).
- By understanding the MODIS Collection 6 format, our GIS backend can combine both MODIS (1 km) and VIIRS (375 m) alerts into one unified real-time map overlay.

4.4 4. Handling Reflective Factory Roofs (False Alarms)

Modern industrial complexes often have large, shiny metal roofs that reflect intense sunlight back to the satellite, creating bright spots. The false-alarm rejection techniques in this paper show us how to distinguish between genuine combustion heat and surface solar reflection.